

YATUKULAN VIVEKANANDAN

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EDUCATION

McMaster University

Expected Graduation: May 2029

Bachelor of Engineering - Computer Engineering Co-op

Hamilton, Ontario

- **Relevant Coursework:** Circuit Analysis, Data Structures & Algorithms, Digital Logic Design, Control Systems

EXPERIENCE

Enedym Inc.

May 2026 – August 2026

Electrical Engineering Intern

Hamilton, Ontario

- Independently developing an in-house **IoT CAN Bus Data Logger**, replacing a \$1000 unit for under **\$500 BOM**
- Performing **PCB layout** in **Altium Designer** integrating CAN transceivers, wireless module, and data storage
- Designing hardware and firmware for real-time **CAN bus** acquisition with **IoT** cloud connectivity
- Building customer-facing **web interface** for streamlined data access, visualization, and remote diagnostics

McMaster Formula Electric

September 2025 – Present

High Voltage Electronics Engineer

Hamilton, Ontario

- Redesigned **Charger Controller** subsystem from requirements derivation through manufacturing and testing
- Integrated **safety logic circuits** and **power converters**; reduced BOM cost **15%** through design optimization
- Performed hands-on **hardware debug** and **electrical design verification** across fail-safe operating states
- Designed **CAN-integrated** charger circuits using **Altium Designer** with comprehensive documentation
- Built hardware test fixtures and conducted **impedance testing** and thermal simulations
- Produced fabrication and assembly documentation, test plans, and technical reports for review

McMaster Chem E Car Team

August 2025 – April 2026

Circuitry Team Member

Hamilton, Ontario

- Reverse-engineered a **HP3435A** multimeter with a failed display system, saving the team **\$150**
- Built an **Arduino**-based data acquisition system to capture voltage and current readings in real time
- Implemented **serial data logging** to Excel for real-time graphing and calibration comparison
- Successfully restored measurement readouts via serial monitor and Excel dashboard
- Analyzed schematics and **datasheets**; conducted **electrical verification** achieving $\pm 1.5\%$ accuracy
- Validated readings against reference equipment with $< 2\%$ deviation

PROJECTS

STM32-Based Mechanical Keyboard | *KiCAD, STM32, USB, C++, Embedded Firmware*

- Designed electrical schematics and **multi-layer PCB** for **82-key** keyboard with **STM32** microcontroller
- Developed **C++ firmware** for real-time matrix scanning; validated timing requirements with logic analyzer
- Performed **schematic and layout reviews** for power management and **USB** interface implementation

SEER – Object & Temperature Detection Device | *Arduino, C/C++, Embedded Systems*

- Developed embedded sensing device for a client with **Usher's Syndrome** to improve environmental navigation
- Debugged firmware and implemented **sensor fusion** algorithm achieving **90%** detection accuracy
- Documented system design and performed hands-on assembly, testing, and validation

ToF Spatial Scanner | *MSP432E401Y, C, Stepper Motor Control, I2C/UART, MATLAB*

- Developed **C** firmware synchronizing a **stepper motor** with **VL53L1X ToF** sensor readings via **I2C**
- Streamed scan data over **UART** to **MATLAB** for real-time 3D point-cloud reconstruction
- Analyzed motor timing, sensor quantization, and UART throughput to characterize scan reliability

TECHNICAL SKILLS

EDA & Design Tools: Altium Designer, KiCAD, LTSpice, PSpice, AutoCAD Electrical, Schematic Capture, Multi-layer PCB Layout

Embedded Systems & Programming: C/C++, Python, MATLAB, Motor Controllers, Embedded Hardware, CAN, SPI, I2C, UART, EtherCAT, USB, Board Bring-up, Hardware Debug, Design Verification

Test, Measurement & Fabrication: Oscilloscopes, Logic Analyzers, Multimeters, Function Generators, Impedance Testing, Datasheet Analysis, Hardware Validation, Soldering